

SINGLE-CONTACT RECOVERY OF INFORMATION METRICS: SHARP THRESHOLDS AND NATIVE REALIZATION

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ABSTRACT. We determine the least ambient dimension in which one contact point identifies a native Hankel information metric for a synchronized quadratic image with $d \geq 2$ parameters. The answer is

$$k_{\min}(d) = d + \left\lceil \frac{3 + \sqrt{16d + 1}}{2} \right\rceil.$$

The observation is a full normal second jet, not a scalar reading. The proof combines the classical maximal-rank theorem for rational curves with a realization theorem: the loading-to-normal-space map is a submersion onto the open family selected by a positive-vector condition. Generic global fibre separation then transfers local polynomial multiplication to the global distance function. We distinguish this exact threshold from the stricter monomial additive-basis problem. We describe the determinantal failure scheme, its tangent map and a native base-point locus, and prove local conditioning estimates. A second realization theorem applies to arbitrary linear matrix families, with an explicit sharp diagonal example. Finally, estimators constructed from the original finite categorical samples give contact and metric recovery bounds; an explicitly observed Poisson exposure experiment permits inference on an open full-dimensional moment set. All earlier contact, endpoint and design results are retained in complete companion sources.

1. INTRODUCTION

Let $M \subset \mathbb{R}^k$ be a residual image and let $R > 0$ be an inverse information matrix. Its contact function is

$$C_R(b) = \inf_{x \in M} (x - b)^\top R^{-1} (x - b). \quad (1.1)$$

At a smooth point, the second jet of this function determines the normal compression of R , and nothing more for an unrestricted ambient matrix. A statistical observation model can, however, confine R to a much smaller family. We ask when this native structure removes the one-point ambiguity.

In the calibrated polynomial experiment considered below,

$$R_\theta = JH(\theta)J^\top, \quad H(\theta)_{ab} = \theta_{a+b}, \quad 0 \leq a, b < k,$$

where J is fixed and invertible and θ belongs to an open finite moment cone of dimension $2k - 1$. A normal vector becomes a polynomial under $J^\top$. The contact data therefore specify a moment functional on products of normal polynomials. For a normal space of dimension $c = k - d$, the necessary inequality is

$$\binom{c+1}{2} \geq 2k - 1. \quad (1.2)$$

The principal result is that this inequality is also sufficient, for every separated root and clock configuration, at a positive globally valid contact point of a generic synchronized loading image.

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Theorem 1.1 (Sharp native threshold). *Fix $d \geq 2$ and the calibrated experiment of Section 3, with $k > d$ root pairs. Put $q_A(\eta) = ((a_i^\top \eta)^2)_{i=1}^k$ for a loading matrix $A \in \mathbb{R}^{k \times d}$. There exist full-column-rank loadings with $q_A(e_1) > 0$ for which the single jet $D^2 C_{R_\theta}(q_A(e_1))$ identifies every θ in the native moment cone if and only if (1.2) holds. When it holds, these conclusions, including validity for the global distance, hold on a nonempty Zariski-open subset of the loading domain where Ae_1 has no zero entries. In particular*

$$k_{\min}(d) = d + \left\lceil (3 + \sqrt{16d + 1})/2 \right\rceil = d + 2\sqrt{d} + O(1). \quad (1.3)$$

At that same contact point unrestricted positive matrices retain an invisible space of dimension $d(2k - d + 1)/2$.

The abstract multiplication theorem used here is classical maximal rank for general rational curves, not a new theorem about their Hilbert functions [1, 2]. The new step is its exact transfer to the native statistical loading family, including positivity and the global distance. Restricting to monomial normal spaces would obscure this transfer: in some dimensions the optimum cannot be attained by any monomial space. Thus the finite additive-basis problem and the native threshold are different problems, not competing formulations of the same optimum.

The failure locus is also structured. It is the rank locus of multiplication on a Grassmannian, rather than a generic matrix determinantal variety. Its base-point subvariety already precludes an unqualified expected-codimension assertion. We give its incidence description and infinitesimal rank map, and obtain a linear distance-to-degeneracy estimate at transverse corank-one points. A general matrix-family realization theorem explains which part of this geometry is specific to Hankel matrices.

We keep statistical observation categories separate. Finite calibrated categorical samples estimate the native information and consequently its contact readings. These readings are correlated functions of the same sample, not independent Gaussian measurements. When exposures are fixed design values, varying them geometrically does not make them unknown likelihood parameters. An explicitly observed Poisson exposure experiment addresses full-dimensional moment inference. The independently noisy contact experiment of the preceding revision is preserved as a different model, with its different offset dependence.

Related work and attribution. Hankel duality with polynomial multiplication is standard [8]. Products of binary-form subspaces also govern Gram spectrahedra; Scheiderer studies product independence and ranks in that setting [10]. Our threshold uses the surjective as well as the injective range of the rational maximal-rank theorem. Neither an injectivity result for a small number of products nor a monomial witness alone supplies the needed surjectivity.

The monomial specialization belongs to the classical finite and restricted additive 2-basis literature [12, 7, 6]. Section 5 makes that specialization exact, gives an elementary economical construction, and proves a strict separation from general subspaces. No optimal restricted-basis constant is claimed. The complete-conormal discriminant argument is classical projective duality [4]; it addresses a different observation category from one prescribed normal block. Parametric asymptotic inference is used in its standard sense [11], and real quantifier elimination remains a separate companion tool [3].

2. EXACT FIBRES OF PRESCRIBED CONTACT DATA

Write $\mathcal{P}_n$ for real polynomials of degree at most n , with coefficient Euclidean norm. Let $n = k - 1$, $p = 2k - 1$, and

$$\ell_\theta \left(\sum_{a=0}^{2n} f_a t^a \right) = \sum_a f_a \theta_a.$$

Let $\Omega \subset \mathbb{R}^p$ be open with $R_\theta = JH(\theta)J^\top > 0$. At prescribed points x_ν assume the global image has one embedded smooth germ and that the local minimizing projection computes (1.1). For parameter differentiation assume analytic germs and a uniform projection tube on compact metric sets. Section 4 verifies these conditions for the native image.

Let Z_ν have orthonormal columns spanning the Euclidean normal space N_ν . Define

$$B_\nu = Z_\nu^\top R_\theta Z_\nu, \quad G_\nu = \frac{1}{2} Z_\nu^\top D^2 C_{R_\theta}(x_\nu) Z_\nu.$$

Lemma 2.1 (Normal block). *One has $G_\nu = B_\nu^{-1}$. The full Hessian has tangent kernel and is determined by this block.*

Proof. The quadratic contact with a manifold equals that with its tangent space T . Minimize $(u - v)^\top R^{-1}(u - v)$ over $v \in T$. At the minimizer, $R^{-1}(u - v)$ is normal, hence $u - v = RZn$. Its normal projection gives $Bn = Z^\top u$. The minimum is $(Z^\top u)^\top B^{-1}Z^\top u$. A local parametrization and the positive tangent Gram matrix justify the minimizing projection by the implicit function theorem and give the same quadratic term. $\square$

Identify $J^\top N_\nu$ with a polynomial subspace $U_\nu \subset \mathcal{P}_n$. The basis-free observation space is the image

$$\mathcal{U} = \text{im} \left(\bigoplus_\nu \text{Sym}^2 U_\nu \xrightarrow{\sum \text{multiplication}} \mathcal{P}_{2n} \right).$$

Only within-contact products occur. Set $\mathcal{L}\theta = (B_\nu(\theta))_\nu$, with direct-sum Frobenius norm.

Theorem 2.2 (Exact fibre for prescribed contacts). *The exact second-jet fibre through θ is*

$$(\theta + \mathcal{U}^\perp) \cap \Omega.$$

Its local dimension is $p - \dim \mathcal{U}$. Recovery on Ω is equivalent to surjectivity of the displayed multiplication map, or to injectivity of $\mathcal{L}$.

Proof. For normal vectors v, w , with coefficient polynomials f_v, f_w , one has

$$v^\top (R_{\theta'} - R_\theta)w = \ell_{\theta' - \theta}(f_v f_w).$$

By Lemma 2.1, equality of the jets is equality of the normal bilinear blocks, hence precisely annihilation of $\mathcal{U}$. Matrix inversion is a diffeomorphism on positive blocks. The linear map to these blocks has rank $\dim \mathcal{U}$, and openness of Ω gives the fibre dimension. $\square$

At one point of a d -manifold, $\dim N = c = k - d$. There are at most $\binom{c+1}{2}$ independent products, proving necessity of (1.2). For unrestricted matrices, fixing the normal-normal block leaves tangent-tangent and mixed blocks free, of total dimension

$$\frac{k(k+1) - c(c+1)}{2} = \frac{d(2k - d + 1)}{2}. \quad (2.1)$$

All sufficiently small such perturbations preserve positivity.

3. THE NATIVE INFORMATION FAMILY

Fix distinct strictly positive rank-one four-cell probability tables M_1, M_2 , roots $0 < r_1 < \dots < r_k < D$, and $m = 2k + 1$ clocks $D < T_1 < \dots < T_m$. In one component use

$$f_1(T) = f_{1,0}(T) \prod_i ((T - r_i - s_i/2)^2 - v_i), \quad p_j = \pi_j M_1 + (1 - \pi_j) M_2,$$

$$\frac{\pi_j}{1 - \pi_j} = \frac{\alpha f_1(T_j)}{(1 - \alpha) f_2(T_j)}.$$

The fixed factors are positive at all clocks. At $s = v = 0$, the free coordinates are log mixing odds and root-sum displacements; the retained variances satisfy $v_i \geq 0$. Exposures $\lambda_j > 0$ multiply independent group information. Put

$$\kappa_j = \pi_j^2 (1 - \pi_j)^2 \sum_{a=1}^4 \frac{(M_{1a} - M_{2a})^2}{p_{ja}}, \quad V_j = (1, -(T_j - r_i)_i^{-1}, -(T_j - r_i)_i^{-2}).$$

The notation lists the k simple poles before the k double poles. Define

$$h(t) = \prod_i (t - r_i)^2, \quad p_*(t) = \prod_i (t - r_i), \quad P(t) = \prod_j (t - T_j),$$

$$d_i = \frac{P(r_i)}{\prod_{a \neq i} (r_i - r_a)^2}, \quad \xi_j = \frac{h(T_j)}{P'(T_j)}, \quad J_{ia} = [t^a] d_i \frac{p_*(t)}{t - r_i}, \quad \rho_j = \frac{\xi_j^2}{p_*(T_j)^2}. \quad (3.1)$$

Proposition 3.1 (Profiled information). *The inverse efficient information in the retained coordinates is*

$$R = JH(\theta)J^\top, \quad \theta = \sum_{j=1}^m \frac{\rho_j}{\lambda_j \kappa_j} w_{2n}(T_j), \quad w_s(t) = (1, t, \dots, t^s)^\top. \quad (3.2)$$

The matrix J is invertible. Independently varying positive exposures gives an open moment cone Ω of dimension $p = 2k - 1$.

Proof. Multiplication by h converts any linear combination of the columns of V to a polynomial of degree at most $2k$. Vanishing at $2k + 1$ clocks implies the zero polynomial; principal parts at each root and the constant at infinity then give zero coefficients. Thus V is invertible. The full information is $V^\top \text{diag}(\lambda_j \kappa_j) V$, and the retained block of its inverse is the inverse nuisance Schur complement.

For arbitrary log-odds increments y_j , interpolate

$$g(t) = \sum_j h(T_j) y_j \frac{P(t)}{(t - T_j) P'(T_j)}.$$

The retained coefficient at r_i is $-g(r_i) / \prod_{a \neq i} (r_i - r_a)^2$. Therefore the retained rows Z_0 of V^{-1} satisfy

$$(Z_0)_{ij} = \frac{\xi_j d_i}{T_j - r_i} = \frac{\xi_j}{p_*(T_j)} (Jw_n(T_j))_i.$$

Substitution into $Z_0 \text{diag}((\lambda_j \kappa_j)^{-1}) Z_0^\top$ proves (3.2). Evaluation at the roots proves independence of the row polynomials of J . The distinct-clock Vandermonde matrix has rank p , so the image of the positive weight orthant is open in $\mathbb{R}^p$. Its Hankel matrices are positive because the clock evaluation vectors span $\mathbb{R}^k$. $\square$

The exposures in this proposition specify information weights, not automatically unknown sampling parameters. Section 8 distinguishes the corresponding statistical experiments.

4. REALIZATION AND THE EXACT THRESHOLD

The next lemma is not restricted to the interpolation matrix (3.1).

Lemma 4.1 (Positive realization and submersion). *Fix any $J \in \text{GL}_k(\mathbb{R})$ and $1 \leq d < k$. A c -plane $U \subset \mathcal{P}_n$, $c = k - d$, is the normal-polynomial space at $q_A(e_1) > 0$ for some full-rank loading A if and only if*

$$JU^\perp \cap \mathbb{R}_{\geq 0}^k \neq \emptyset. \quad (4.1)$$

Here the assertion concerns the local immersed image. The spaces satisfying (4.1) form a nonempty Euclidean-open subset of $\text{Gr}(c, k)$. On the domain where A has full column rank and Ae_1 has no zero coordinate, the map

$$A \longmapsto U(A) = J^\top (\text{ran}(\text{diag}(Ae_1)A))^\perp$$

is a submersion. In particular its image is Zariski dense.

Proof. Put $z = Ae_1$. The tangent space is $T = \text{ran}(\text{diag}(z)A)$, which contains $z^2 > 0$. Since $U = J^\top T^\perp$, taking orthogonal complements gives $T = JU^\perp$; this proves necessity.

Conversely choose $t > 0$ in $T = JU^\perp$ and extend it to a basis $t, t_2, \dots, t_d$ of T . Put $z_i = \sqrt{t_i}$ and

$$A = [z, \text{diag}(z)^{-1}t_2, \dots, \text{diag}(z)^{-1}t_d].$$

Then $\text{diag}(z)A$ has precisely that basis, A has rank d , and $q_A(e_1) = t$. The positive-vector condition persists when a basis of T is perturbed; it is nonempty, for example for planes containing $\mathbf{1}$.

More precisely, on each coordinate-sign chamber for z , the map $A \mapsto B = \text{diag}(z)A$ is a diffeomorphism onto the full-rank matrices whose first column is strictly positive. Its inverse is $z_i = \pm\sqrt{B_{i1}}$, with the fixed chamber signs, and $A_{i\ell} = B_{i\ell}/z_i$. The map $B \mapsto \text{ran } B$ is the usual frame-bundle submersion. Orthogonal complement and the fixed J transformation preserve this property. A nonempty real open subset of a real Grassmannian is Zariski dense, as seen in any affine matrix chart. $\square$

Lemma 4.2 (Global separation and distance). *For $k > d$, a nonempty Zariski-open set of full-rank A with Ae_1 nowhere zero satisfies*

$$q_A^{-1}(q_A(e_1)) = \{e_1, -e_1\}.$$

For each such loading the global image has one embedded analytic germ at $b_0 = q_A(e_1)$. The global distance agrees with the distance to that germ in a neighbourhood of b_0 , uniformly for R in any compact positive metric set.

Proof. Equality of coordinatewise squares means $A\eta = D_s Ae_1$ for a sign vector $s \in \{\pm 1\}^k$. For each nonconstant s , the condition $D_s Ae_1 \in \text{ran } A$ is algebraic, given by augmented $(d+1)$ -minors. It is proper: take z with no zero entries, so $z, D_s z$ are independent, and extend z to a d -plane avoiding $D_s z$. Excluding these finitely many proper sets leaves only the two constant signs, which give $\pm e_1$.

Full rank gives $\|q_A(\eta)\|_2 \geq k^{-1/2}\|A\eta\|_2^2 \geq a\|\eta\|_2^2$, so q_A is proper. Its derivative has rank d at both preimages; their local images agree by evenness. Properness excludes any other sheet accumulating at b_0 . Thus one embedded analytic germ represents the global image there. For $b \rightarrow b_0$, the trial image b_0 bounds the minimum by $C\|b - b_0\|^2$. Uniform positive metric bounds force every minimizing image to tend to b_0 . The germ therefore supplies the global minimum, and its positive tangent Hessian supplies the analytic projection on a uniform smaller tube. $\square$

Proposition 4.3 (Classical rational maximal rank). *For $4 \leq c \leq n+1$, a general complex c -plane $U \subset H^0(\mathbb{P}^1, \mathcal{O}(n))$ has*

$$\text{rank}(\text{Sym}^2 U \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(2n))) = \min\left\{\binom{c+1}{2}, 2n+1\right\}. \quad (4.2)$$

The same nonempty Zariski-open rank locus exists over $\mathbb{R}$.

Reduction to the classical theorem. A general linear series of this dimension is base-point-free and embeds the rational curve in $\mathbb{P}^{c-1}$: equivalently a general projection of the rational normal curve avoids its secant and tangent locus when the target dimension is at least three. The resulting degree- n , genus-zero curve is nonspecial, since $H^1(\mathbb{P}^1, \mathcal{O}(n)) = 0$. The maximal-rank theorem for general nonspecial curves [1, 2], applied to restriction of quadrics, gives (4.2). That restriction

is exactly multiplication of the c sections defining the embedding. This paragraph applies the classical theorem; it does not reprove it. Maximal minors have real coefficients in Grassmannian charts, and a nonzero such polynomial cannot vanish on every real chart point. This gives the real assertion. The complete-series case is also immediate by monomial multiplication. $\square$

Proof of Theorem 1.1. Necessity follows from Section 2. If (1.2) holds and $d \geq 2$, then $c \geq 5$. Proposition 4.3 gives a nonempty Zariski-open full-product locus in $\text{Gr}(c, k)$. It intersects the nonempty real open realizable locus of Lemma 4.1. Hence its pullback to loadings is nonempty. The pullback is Zariski open on the rank domain: normal frames are rational functions of loading entries on pivot charts, and the condition is nonvanishing of a maximal minor. The loading domain is irreducible, so this open set is dense. Intersect it with Lemma 4.2's dense open set. Theorem 2.2 now supplies native recovery from the global distance, for every positive exposure vector with the same J .

Writing $k = d + c$ in (1.2) gives $c^2 - 3c - 4d + 2 \geq 0$, whose positive root is $(3 + \sqrt{16d + 1})/2$. This proves (1.3); (2.1) supplies the remaining assertion. $\square$

Remark 4.4. This argument proves genericity in the loading family, not merely openness around a selected witness. It does not assert that every locally realizable U has a globally separated realization with exactly the same U . The two open sets are intersected in loading space. For fixed real roots and clocks the resulting open set contains rational A , without requiring J itself to have rational entries.

For $d = 2, 3, 4, 5, 6$ the optimal k are respectively 7, 8, 10, 11, 13. At the optimum the full-symmetric measurement $S \mapsto (a_i^\top S a_i)_i$ is necessarily noninjective for every $d \geq 5$: check $d = 5, 6$ directly; for $d \geq 7$, (1.2) already holds with $c = d$, so $k_{\min} \leq 2d < d(d + 1)/2$. Thus the theorem does not hide a full-symmetric phase-retrieval assumption.

5. MONOMIAL SPACES AND RESTRICTED ADDITIVE BASES

For $S \subset \{0, \dots, n\}$ let $U_S = \text{span}\{t^a : a \in S\}$. Then

$$\text{span}(U_S U_S) = \mathcal{P}_{2n} \iff S + S = \{0, \dots, 2n\}. \quad (5.1)$$

This is exactly a restricted additive 2-basis problem: the largest basis element is half the represented range [6]. General finite-basis results [12, 7] provide context but cannot be substituted without checking that restriction.

Proposition 5.1 (A strict monomial obstruction). *If $n \geq 3$ and $S + S = \{0, \dots, 2n\}$, then*

$$\binom{|S| + 1}{2} \geq 2n + 2.$$

Consequently at $(d, k) = (3, 8)$ and $(5, 11)$ no monomial normal space attains the native optimum, whereas generic polynomial spaces do.

Proof. Representation of $0, 1, 2n - 1, 2n$ forces $0, 1, n - 1, n \in S$. The unordered pairs $(0, n)$ and $(1, n - 1)$ are distinct for $n \geq 3$ and have equal sum. Thus at least one of the $\binom{|S| + 1}{2}$ pairs is redundant when representing $2n + 1$ distinct integers. At the two displayed optima the number of pairs is exactly $2n + 1$ (15 and 21, respectively), contradicting the strengthened bound. $\square$

Proposition 5.2 (An elementary economical monomial family). *For $a \geq 2$ and $n \geq 2a$, write $n = qa + r$, $0 \leq r < a$, and set*

$$S = [0, a - 1] \cup \{a, 2a, \dots, qa\} \cup [n - a + 1, n],$$

where intervals denote integer sets. Then $S + S = [0, 2n]$ and $|S| \leq 2a + \lfloor n/a \rfloor - 1$. In particular there are such bases with $|S| \leq 2\sqrt{2n} + O(1)$.

Proof. The sums of $[0, a-1]$ with $0, a, \dots, qa$ cover $[0, qa+a-1]$, which contains $[0, n]$. The sums of $[n-a+1, n]$ with the same multiples cover $[n-a+1, n+qa] = [n-a+1, 2n-r]$. Finally the upper interval added to itself covers $[2n-2a+2, 2n]$, overlapping the previous range since $r \leq a-1$. These intervals cover $[0, 2n]$. The last multiple qa belongs to the upper interval, giving the cardinality bound. Choose a to be a nearest integer to $\sqrt{n/2}$; the finitely many smaller n can be handled by the full interval. $\square$

Padding a basis by additional exponents preserves coverage. The earlier $k = 3d+2$ construction corresponds to $S = [0, d] \cup [2d+1, 3d+1]$ and remains an explicit elementary witness. The referee's smaller sets $\{0, 1, 3, 5, 6\}$ and $\{0, 1, 2, 5, 8, 11, 12, 13\}$ are valid restricted bases. Neither is needed for the exact unrestricted threshold. Since $d \geq 1$ leaves a tangent exponent, the native interpolation matrix realizes any such monomial complement locally: each column of J has one strict coordinatewise sign, so a signed column gives the positive vector required in (4.1). Generic perturbation then gives the global conclusion without preserving monomiality.

6. FAILURE GEOMETRY AND CONDITIONING

For algebraic statements in this section we complexify and homogenize to binary forms $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$; metric statements concern the real points of the resulting loci. Dehomogenization identifies this with $\mathcal{P}_n$ and preserves multiplication. On $\text{Gr}(c, V_n)$ let $\mathcal{S}$ be the tautological bundle, and consider

$$\mu : \text{Sym}^2 \mathcal{S} \longrightarrow V_{2n} \otimes \mathcal{O}, \quad q = \binom{c+1}{2}, \quad p = 2n+1.$$

Assume $q \geq p$. Let Δ_r be the scheme cut out by $(r+1)$ -minors of μ and $\Delta = \Delta_{p-1}$.

Proposition 6.1 (Incidence, tangent map, and fibres). *Set-theoretically, Δ is the projection of*

$$\{(U, [\ell]) \in \text{Gr}(c, V_n) \times \mathbb{P}(V_{2n}^*) : \ell(fg) = 0 \text{ for all } f, g \in U\}.$$

Equivalently U is totally isotropic for a nonzero Hankel bilinear form. At a rank- r point U , put $K_U = \ker \mu_U$ and $Q_U = V_{2n}/U^2$. The differential controlling the determinantal scheme is

$$\begin{aligned} \beta_U : \text{Hom}(U, V_n/U) &\longrightarrow \text{Hom}(K_U, Q_U), \\ \beta_U(\phi) \left(\sum_i f_i \odot g_i \right) &= \sum_i \{\phi(f_i)g_i + f_i\phi(g_i)\} \pmod{U^2}. \end{aligned} \tag{6.1}$$

The Zariski tangent space to Δ_r is $\ker \beta_U$. If β_U is onto, Δ_r is smooth there of codimension $(q-r)(p-r)$. On the corresponding native rank stratum the exact contact fibre has dimension $p-r$.

Proof. Rank failure is equivalent to a nonzero functional annihilating the multiplication image. Projectivity of the second factor makes the incidence projection closed. Its bilinear matrix in monomial coordinates is Hankel. To differentiate, write nearby spaces as graphs of maps $U \rightarrow V_n/U$. Differentiation of a product gives (6.1); changing a chosen lift adds an element of U^2 , so it is well defined. Near a nonzero r -minor, invertible row and column operations reduce the matrix to an invertible r -block and its Schur complement. The rank- $\leq r$ equations are precisely the entries of that complement. Their differential is the induced kernel-to-cokernel map β_U . The tangent and smoothness assertions follow from these equations and the submersion theorem. The fibre statement follows from Theorem 2.2 and Lemma 4.1. $\square$

The tangent assertion concerns the determinantal scheme defined by the minors; it is not an assertion that every rank locus is reduced or has expected codimension.

Proposition 6.2 (A native base-point locus). *Suppose $2 \leq c \leq n$. The subvariety $\mathcal{B}$ of spaces whose forms have a common projective zero is irreducible of codimension $c - 1$ in $\text{Gr}(c, n + 1)$ and is contained in Δ_{p-2} . It meets the real native realizable open set for every separated root and clock configuration. Its loading pullback has the same local codimension along its smooth points.*

Proof. For a fixed linear factor l , the spaces are $\text{Gr}(c, lV_{n-1}) \cong \text{Gr}(c, n)$. The incidence over $\mathbb{P}^1$ is irreducible of dimension $1 + c(n - c)$. A general member has exactly that one simple common factor: choose the residual space with no common zero and with some form not vanishing at the specified point. Thus the projection is generically one-to-one and its codimension is $c(n + 1 - c) - 1 - c(n - c) = c - 1$. Products are divisible by l^2 , so their span has dimension at most $\dim V_{2n-2} = 2n - 1 = p - 2$.

Choose a real common zero $t > D$. For $U \subset \{f : f(t) = 0\}$, its coefficient orthogonal complement contains $w_n(t)$. Formula (3.1) gives

$$(Jw_n(t))_i = d_i \frac{p_*(t)}{t - r_i} < 0.$$

Its negative is strictly positive, proving native local realizability by Lemma 4.1. The codimension is preserved under its submersion at smooth points. $\square$

In particular, whenever the generic-matrix expected codimension $q - p + 1$ exceeds $c - 1$, the whole failure locus cannot have that expected codimension. We do not replace this explicit component-sized family and tangent description by an unsupported classification of every irreducible component.

Fix coefficient norms and Euclidean normal norms. Let γ be the smallest singular value of $\mathcal{L}$. Positive singular values of multiplication and its dual agree; passage from an orthonormal polynomial frame to $J^\top Z$ has uniformly bounded condition number for fixed J . Thus their smallest singular values are comparable.

Theorem 6.3 (Stable inversion and transverse degeneracy). *If $m_0 I \leq B_\nu \leq M_0 I$ and $\hat{G}_\nu = G_\nu + E_\nu$ with $\max \|E_\nu\|_{\text{op}} \leq (2M_0)^{-1}$, then*

$$\left\| \mathcal{L}^\dagger(\hat{G}_\nu^{-1})_\nu - \theta \right\|_2 \leq \frac{2M_0^2}{\gamma} \left(\sum_\nu \|E_\nu\|_F^2 \right)^{1/2}. \quad (6.2)$$

On the Grassmannian, the least multiplication singular value is at most a fixed constant times distance to Δ . Near a real rank- $(p - 1)$ point where (6.1) is onto, it is comparable above and below to that distance. The same local comparison holds in loading space for distance to the pulled-back failure locus.

Proof. The inverse identity gives $\hat{G}_\nu^{-1} - B_\nu = -\hat{G}_\nu^{-1} E_\nu B_\nu$; the assumed bounds imply its Frobenius norm is at most $2M_0^2 \|E_\nu\|_F$. Apply $\mathcal{L}^\dagger$, of norm γ^{-1} .

Multiplication expressed in orthonormal frames varies Lipschitzly on finitely many compact frame charts. Singular values are Lipschitz in the matrix entries and vanish on Δ , proving the upper estimate. At corank one the Schur complement used above is a $1 \times (q - p + 1)$ row $s(U)$. Bounded invertible block operations show that the least singular value is comparable to $\|s(U)\|$. Surjectivity of $\beta_U = Ds(U)$ gives local coordinates in which s is a coordinate projection; its norm is comparable to distance to $s^{-1}(0)$. The loading submersion and the frame norm comparisons give the final assertion. $\square$

This does not assert linear distance control at singular or nontransverse points. Compact sets separated from Δ have a uniform positive stability constant; that assertion follows by continuity, without a dimension-free numerical lower bound.

7. TRANSFER TO OTHER STRUCTURED FAMILIES

Let $\mathcal{K} \subset \text{Sym}_k$ be any fixed s -dimensional linear space containing a positive matrix. For a c -plane N , consider compression $X \mapsto X|_{N \times N}$ and let $\rho(\mathcal{K}, c)$ be its maximum rank over the Grassmannian.

Theorem 7.1 (Structured-family realization). *For a generic full-rank loading with Ae_1 nowhere zero, the normal compression has rank $\rho(\mathcal{K}, c)$, and the global contact fibre within any open positive subset of $\mathcal{K}$ has dimension $s - \rho(\mathcal{K}, c)$. In particular every generically injective normal compression problem has a positive globally valid synchronized realization. For the diagonal family, generic injectivity is equivalent to*

$$\binom{c+1}{2} \geq k.$$

Its least k at fixed $d = k - c$ is $d + \lceil (1 + \sqrt{1 + 8d})/2 \rceil$.

Proof. Maximal compression rank is a nonempty Zariski-open condition on the Grassmannian. Lemma 4.1 with $J = I$ and Lemma 4.2 transfer it to the loading domain. The inverse normal block has the same fibres as compression, so ordinary rank-nullity proves the fibre assertion.

For diagonal matrices, write the rows of a normal frame as $u_i^\top \in \mathbb{R}^c$. Compression is $\sum_i \theta_i u_i u_i^\top$. The c vectors e_a and the $c(c-1)/2$ vectors $e_a + e_b$, $a < b$, have linearly independent outer products: off-diagonal entries first force the coefficients of the sums to vanish and diagonal entries then force the others to vanish. Selecting the first k of these vectors, starting with all e_a , gives a full-column-rank frame whenever $c \leq k \leq \binom{c+1}{2}$. For larger k , retain the complete family and add arbitrary rows; the rank is the maximum possible. This proves generic rank $\min(k, \binom{c+1}{2})$. Substituting $k = d + c$ gives $c^2 - c - 2d \geq 0$ and the formula. $\square$

The diagonal example is itself an information family: independent Bernoulli log-odds coordinates have inverse information $\text{diag}((\lambda_i p_i (1 - p_i))^{-1})$. Independent positive exposures give the full positive diagonal cone. The general theorem does not assert that every linear matrix family is generated by the polynomial experiment; it establishes the common realization mechanism and gives a second genuine observation model.

8. RECOVERY FROM FINITE CALIBRATED SAMPLES

Keep the roots, clocks and calibrated tables fixed. Set $\delta = M_1 - M_2 \neq 0$ and assume $\pi_j \in [\varepsilon, 1 - \varepsilon]$, $0 < \varepsilon < 1/2$. On this interval define

$$\kappa(u) = u^2(1-u)^2 \sum_{a=1}^4 \frac{\delta_a^2}{M_{2a} + u\delta_a}, \quad g(u) = 1/\kappa(u).$$

The denominators and κ are uniformly positive, so g has a finite Lipschitz constant L_g . These assumptions hold on any compact interior calibrated parameter set.

Theorem 8.1 (Fixed-design categorical recovery). *At clock j , observe $n_j \geq 1$ independent categorical samples of law p_j , independently across groups. Exposures $\lambda_j > 0$ are known design weights. If $\bar{p}_j$ is the empirical four-cell frequency, set*

$$\hat{\pi}_j = \text{proj}_{[\varepsilon, 1-\varepsilon]} \frac{\langle \bar{p}_j - M_2, \delta \rangle}{\|\delta\|_2^2}, \quad W_{\cdot j} = \frac{\rho_j}{\lambda_j} w_{2n}(T_j), \quad \hat{\theta} = W(g(\hat{\pi}_j))_j.$$

Then $\hat{\theta} \in \Omega$ and

$$\mathbb{E} \|\hat{\theta} - \theta\|_2^2 \leq \frac{\|W\|_{\text{op}}^2 L_g^2}{\|\delta\|_2^2} \sum_j n_j^{-1}. \quad (8.1)$$

On fixed compact design sets, the induced estimates $\hat{R} = JH(\hat{\theta})J^\top$ and $\hat{G}_\nu = (Z_\nu^\top \hat{R} Z_\nu)^{-1}$ have mean squared error $O(\sum_j n_j^{-1})$. If $\mathcal{L}$ is injective, inversion of these induced jets returns $\hat{\theta}$ exactly.

Proof. Before projection the estimator of π_j is unbiased, since $p_j - M_2 = \pi_j \delta$. For one one-hot cell vector X , $\text{Var}(\langle X, \delta \rangle) \leq \mathbb{E}\langle X, \delta \rangle^2 \leq \|\delta\|_2^2$. Projection onto the interval cannot increase distance to the true π_j , so $\mathbb{E}|\hat{\pi}_j - \pi_j|^2 \leq (n_j \|\delta\|_2^2)^{-1}$. Apply the Lipschitz bound for g and the operator norm of W to obtain (8.1). All estimated weights in (3.2) are positive. On a compact design set they range in a fixed positive box, whose moment image is a compact convex subset of Ω . The linear map $\theta \mapsto R$ and normal-block inversion are uniformly Lipschitz there. Finally $\hat{G}_\nu^{-1} = \mathcal{L}_\nu \hat{\theta}$, so $\mathcal{L}^\dagger(\hat{G}_\nu^{-1})_\nu = \hat{\theta}$. $\square$

The bound allows the π_j to be coupled through the original polynomial model. It neither assumes independent free mixing parameters nor asserts a p -dimensional fixed-design likelihood. When $n_j/N \rightarrow \lambda_j > 0$, it is an $O(N^{-1})$ bound for the information metric normalized by that common scale; one may also use the known actual weights n_j/N and target their corresponding finite-design metric.

Theorem 8.2 (Observed exposures and full-dimensional inference). *Consider the following additional sampling protocol. Observe independent counts $N_j \sim \text{Pois}(N\lambda_j)$ and, conditional on them, N_j categorical marks of the same calibrated law p_j . Assume $0 < \lambda_- \leq \lambda_j \leq \lambda_+$. Estimate λ_j by projecting N_j/N onto this interval, and estimate π_j as above, using any value in $[\varepsilon, 1 - \varepsilon]$ when $N_j = 0$. Substitution into (3.2) gives an estimator with uniform squared risk at most K/N , where K depends only on the fixed compact design bounds.*

The image of an open exposure box is open in the p -dimensional moment cone. If the π_j are known and the exposures are interior, the estimator has limiting covariance

$$\Sigma_\theta = \sum_j \frac{\rho_j^2}{\lambda_j^3 \kappa_j^2} w_{2n}(T_j) w_{2n}(T_j)^\top > 0 \quad (8.2)$$

after multiplication of its error by $\sqrt{N}$. In this known-mark-law subexperiment the N^{-1} squared-risk rate is locally unimprovable and the covariance is the efficient covariance for the moment functional.

Proof. Projection gives $\mathbb{E}|\hat{\lambda}_j - \lambda_j|^2 \leq \lambda_+/N$. If $X \sim \text{Pois}(a)$, then

$$\mathbb{E}[X^{-1}; X > 0] \leq 2\mathbb{E}[(X+1)^{-1}] = 2(1 - e^{-a})/a, \quad \Pr(X = 0) = e^{-a} \leq 1/a.$$

Conditional application of Theorem 8.1 therefore bounds the mixing-estimation risk by a fixed constant divided by $N\lambda_-$. The functions $(\lambda, u) \mapsto \rho_j/(\lambda\kappa(u))$ are uniformly Lipschitz on the compact rectangle. Summing their error bounds and applying the fixed moment matrix proves the risk assertion.

For known π , the Jacobian of the weight-to-moment map in exposures has columns $-\rho_j w_{2n}(T_j)/(\lambda_j^2 \kappa_j)$ and has rank p . The submersion theorem proves the open-image assertion. Independent Poisson central limit theorems and the delta method give (8.2); the Vandermonde rank makes it positive definite. The exposure information per unit N is $\text{diag}(1/\lambda_j)$, and applying this functional Jacobian to its inverse gives exactly (8.2). The regular Poisson likelihood is locally asymptotically normal, so the usual local asymptotic minimax bound for smooth functionals applies [11, Chapters 7–8]. Alternatively a single exposure direction with nonzero moment derivative and two $N^{-1/2}$ -separated alternatives gives the order lower bound. No new minimax principle is being asserted. $\square$

Counts are part of the observation in Theorem 8.2. With fixed known group sizes, unknown exposures cannot be inferred from the conditional mark law alone. This distinction is necessary for a genuine full-cone statistical statement.

8.1. Finite contact readings and the joint scaling. Work at a globally separated informative point b_0 . Let Z be its orthonormal normal frame. Use the q vectors v_a consisting of its columns and their pairwise sums. Quadratic evaluation gives an isomorphism $\mathcal{Q} : \text{Sym}_c \rightarrow \mathbb{R}^q$. Define exact symmetric contact readings

$$\mu_h(\theta)_a = \frac{1}{2}\{C_{R_\theta}(b_0 + hv_a) + C_{R_\theta}(b_0 - hv_a)\}.$$

Fix a compact convex $\Theta \subset \Omega$ containing the true parameter and the clipped plug-in estimates above. Reduce $h_0 > 0$ so all targets are positive and are in a common analytic tube. Uniform symmetric Taylor expansion, including a parameter derivative, gives

$$h^{-2}\mu_h(\vartheta) = \mathcal{Q}((\mathcal{L}\vartheta)^{-1}) + h^2r_h(\vartheta), \quad \sup_{0 < h \leq h_0} \|r_h\|_{C^1(\Theta)} < \infty. \quad (8.3)$$

Here $\mathcal{L}$ is the single normal block. The factor follows from $G = \frac{1}{2}D^2C|_N$.

Theorem 8.3 (Raw samples to finite-offset recovery). *Form $\hat{\mu}_h = \mu_h(\hat{\theta})$ from the categorical or observed-exposure estimator, and define*

$$\tilde{\theta}_h = \mathcal{L}^\dagger \left[\mathcal{Q}^{-1}(h^{-2}\hat{\mu}_h) \right]^{-1}.$$

For small enough h all inverses exist and, if the raw estimator has squared risk at most K/N ,

$$\mathbb{E}\|\tilde{\theta}_h - \theta\|^2 \leq 2K/N + 2C^2h^4/\gamma^2. \quad (8.4)$$

For fixed design, $h_N = o(N^{-1/4})$ makes this reconstruction asymptotically equivalent at scale $\sqrt{N}$ to the raw estimator. Along a sequence of designs with uniform tube and metric constants, it suffices that $\sqrt{N}h_N^2/\gamma_N \rightarrow 0$. Deterministic contact-computation error of Euclidean size at most τ_N adds parameter error at most $C'\tau_N/(\gamma_N h_N^2)$, provided this normalized error is small enough for positive inversion.

Proof. On the compact positive-block region, (8.3) and smooth matrix inversion give

$$F_h(\vartheta) := [\mathcal{Q}^{-1}(h^{-2}\mu_h(\vartheta))]^{-1} = \mathcal{L}\vartheta + h^2E_h(\vartheta), \quad \|E_h\|_{C^1} \leq C.$$

Thus $\tilde{\theta}_h = \hat{\theta} + h^2\mathcal{L}^\dagger E_h(\hat{\theta})$. The norm of the second term is at most Ch^2/γ , which proves the risk and scaling assertions. Perturbing the readings before inversion by τ_N perturbs its matrix argument by at most $\|\mathcal{Q}^{-1}\|\tau_N/h_N^2$. The inverse identity used in Theorem 6.3, followed by $\mathcal{L}^\dagger$, proves the error bound. $\square$

This is a two-stage procedure from raw observations. It uses a fitted calibrated model to compute the finite contact functions; the contact readings are not separately supplied by an oracle. Its covariance is correlated and range-compatible, explaining why the raw term in (8.4) is not multiplied by h^{-4} . Direct use of $\hat{\theta}$ is more economical when the entire fitted model is retained; the two-stage result quantifies a pipeline that retains only finitely many contact readings.

APPENDIX A. THE INDEPENDENT NOISY-CONTACT EXPERIMENT

For comparison, retain the distinct experiment $Y_i = \mu_h(\theta) + \sigma\xi_i$, $\xi_i \sim N(0, I_q)$ independent. The exact finite-offset mean, not its Taylor truncation, is fitted. For compact convex Θ and sufficiently small fixed h , there are $a, b > 0$ such that

$$ah^2\|\theta - \theta'\| \leq \|\mu_h(\theta) - \mu_h(\theta')\| \leq bh^2\|\theta - \theta'\|.$$

Indeed $F_h = \mathcal{L}\theta + h^2E_h$ above and $h^2\|DE_h\| \leq \gamma/2$ imply a global lower Lipschitz bound for F_h by integration along segments in Θ ; positive-block inversion and $\mathcal{Q}$ transfer it to μ_h . The upper bound is immediate from (8.3). Least squares therefore has

$$\sup_{\theta \in \Theta} \mathbb{E}\|\hat{\theta}_N - \theta\|^2 \leq 4q\sigma^2/(Na^2h^4).$$

To see this, its fitted mean is at distance at most twice the sample-mean error from the true mean. The expected squared sample-mean error is $q\sigma^2/N$.

At an interior parameter the usual likelihood Taylor expansion gives information $I_h = \sigma^{-2} D\mu_h^\top D\mu_h > 0$. The least-squares normal equations yield asymptotic covariance I_h^{-1} ; Gaussian tails and the global bound justify convergence of local risks. The local asymptotic min-max bound is $\text{tr}(I_h^{-1})/N$ [11]. Dividing I_h by h^4 and using (8.3) gives the Gram matrix of $\delta \mapsto -\sigma^{-1} \mathcal{Q}(B^{-1}(\mathcal{L}\delta)B^{-1})$. Its least eigenvalue is comparable to γ^2/σ^2 on fixed metric bounds. These statements retain the preceding revision's noisy theorem without confusing it with Section 8. A quadratic-truncated fit at fixed h would generally have a nonvanishing $O(h^2)$ bias and is not used here.

APPENDIX B. EXPLICIT COORDINATES AND THE EARLIER CONSTRUCTIVE FAMILY

For any nonorthonormal normal frame $W = ZO$, set $K = W^\top W$ and $G = \frac{1}{2}W^\top D^2 C_R W$. The correct dual readout is

$$KG^{-1}K = W^\top RW.$$

In fact $K = O^\top O$ and $G = O^\top B^{-1}O$, and direct inversion proves the identity. If W has columns $J^{-\top}e_a$ for $a \in S$, its readout entries are θ_{a+b} . This preserves the normalization needed by explicit monomial reconstruction.

For the old family $k = 3d + 2$, take $E = [d + 1, 2d]$ and $S = [0, d] \cup [2d + 1, 3d + 1]$. The column $-Je_{d+1}$ is positive. Choosing it as t in Lemma 4.1 gives the explicit loading of revision 109. Its three sum intervals are $[0, 2d]$, $[2d + 1, 4d + 1]$, and $[4d + 2, 6d + 2]$. It therefore remains a full-product witness, with unrestricted fibre dimension $5d(d + 1)/2$. Its complete original proof and exact examples remain unchanged in the v109 companion. Theorem 1.1 strengthens, rather than retracts, that result.

APPENDIX C. RETAINED COMPLETE-CONTACT, DESIGN, AND ENDPOINT RESULTS

The complete v109 manuscript and the complete v108 companion, including its materialized inputs, remain part of this revision package. Their proofs are not deleted or silently transferred to weaker hypotheses. We record the precise interfaces.

For injective $\mathcal{A}_A : \text{Sym}_d \rightarrow \mathbb{R}^k$, the complete conormal union is dense in the inverse image under $\mathcal{A}_A^*$ of singular symmetric matrices. Its quadratic vanishing space is the determinant line for $d = 2$ and zero for $d \geq 3$. One elementary verification diagonalizes a symmetric matrix: every quadratic monomial in at least three diagonal variables omits a variable, so vanishing whenever one variable is zero forces all coefficients to vanish. In dimension two substitution of rank-one matrices leaves precisely a multiple of the determinant. Kernel and mixed terms in the pullback vanish because singular symmetric matrices span Sym_d . This is the coordinate form of the classical Veronese/discriminant result [4], not the source of the single-contact theorem. The two-dimensional ambiguity has $B_{ii} = 0$, $B_{ij} = \det(a_i, a_j)^2$ and is intersected with the native positive cone. In root coordinates the native linear span is characterized by

$$\frac{(r_j - r_i)X_{ij}}{d_i d_j} - \frac{(r_l - r_i)X_{il}}{d_i d_l} + \frac{(r_l - r_j)X_{jl}}{d_j d_l} = 0 \quad (i < j < l).$$

These relations leave k diagonal and $k - 1$ divided-difference coordinates. Testing B gives the preserved two-parameter transversality criterion. Equality of second jets is not equality of all higher contacts.

The deterministic exact-real scalar dual-jet oracle still requires exactly $p = 2k - 1$ queries under full product rank. Squares span products by polarization, so a dual basis gives sufficiency. Along the transcript at an interior parameter, fewer scalar linear queries leave a nonzero common

kernel; small perturbations along it preserve every adaptive response, giving necessity. This is not a statistical sample bound.

For reciprocal design weights $\lambda_j = a_j/\omega_j$, $a_j = \rho_j/\kappa_j > 0$, let V_m be the moment matrix and

$$\chi(\theta) = \min_{V_m \omega = \theta, \omega > 0} \sum_j a_j / \omega_j.$$

The zeroth moment bounds the fibre; the objective blows up at its boundary and is strictly convex. It has a unique minimizer, and its diagonal Hessian $D > 0$ gives $D^2\chi = (V_m D^{-1} V_m^\top)^{-1} > 0$ by constrained differentiation. With $m = p + 2$, the fibre has dimension two; every ray from its minimizer meets a higher level once, giving circular level sets. The unit-budget moment image is $\{\chi \leq 1\}$, with singleton boundary and circular interior design fibres. Homogeneity $\chi(t\theta) = t^{-1}\chi(\theta)$ makes the boundary regular relative to the open cone. Injective contact recovery transports this geometry to observed coordinates.

For a labelled algebraic semialgebraic value graph, fixed-dimensional endpoint representations

$$F(x) = \min_{z \geq 0} (x, z)^\top \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} (x, z)$$

with a normalized positive block matrix are described exactly by positivity, normalization, and the quantified KKT relations $z \geq 0$, $Cz + B^\top x \geq 0$, $z_i(Cz + B^\top x)_i = 0$, and the value equality. Strict convexity in z makes these necessary and sufficient. Real quantifier elimination gives the representation fibre at fixed hidden dimension [3]. Enumeration over dimension terminates under a finite-representability promise; no unconditional dimension bound follows. The explicit visible and first-degenerate-stratum results remain in the full companion. This value-graph input differs from optimizer-map inverse parametric programming [5, 9].

Finally the native local root reduction is unchanged. The expansion $p(u, v) = p_0 + L_p u + D_p v + O((\|u\| + \|v\|)^2)$ has the Fisher matrix of Proposition 3.1. A target tb , synchronized $v = tq_A(\eta)$ and free displacement tu give, after profiling and division by t^2 , $(q_A(\eta) - b)^\top R^{-1}(q_A(\eta) - b)$. Full rank of A gives coercivity and bounded scaled minimizers, so Taylor errors are uniform. Physical splitting has size $\sqrt{t}$. For C^2 nonlinear loadings $g_i(0) = 0$, $Dg_i(0) = a_i^\top$, one has $g_i(\sqrt{t}\eta)^2 = t(a_i^\top \eta)^2 + O(t^{3/2})$ on bounded sets. These local statements do not require injectivity on all of Sym_d ; global chart selection remains governed by the inherited finite-sheet hypotheses.

APPENDIX D. PROOF DEPENDENCIES AND REPRODUCIBILITY

The principal dependency chain is normal-block duality, native score interpolation, the loading submersion, classical rational maximal rank, and global fibre separation. The determinantal and finite-sample theorems then use only the stated compactness and rank assumptions. The full component structure of every multiplication rank locus is not needed by the sharp threshold, and is not asserted here. Transverse conditioning is explicitly conditional on the displayed differential being onto.

Revision 110 is based on the v109 review at commit `87b63e155dac402ec6a9d0a34f69ff0ba5363fdb`.

The previously reviewed principal source is at commit `60999f38f745072d240909fe8c77aee5be2df69a`.

All inherited paths are retained. The response and dependency documents identify the classical theorem separately from the new native transfer and distinguish a source-bound build receipt from a queued workflow. Finite exact diagnostics verify specified instances and identities; they do not certify universal proofs or journal-level significance.

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